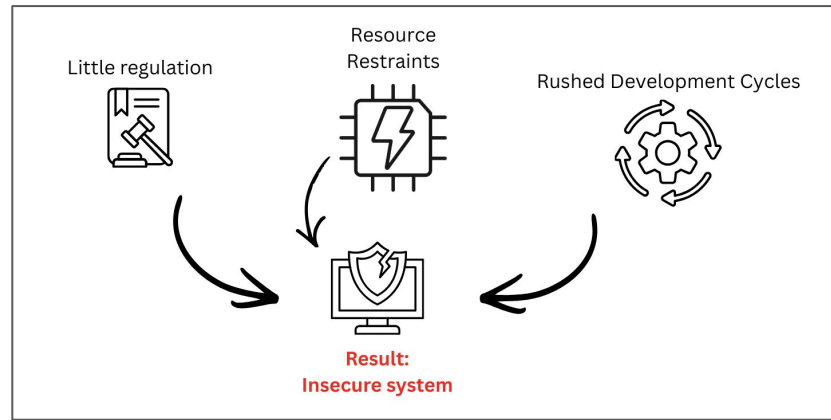
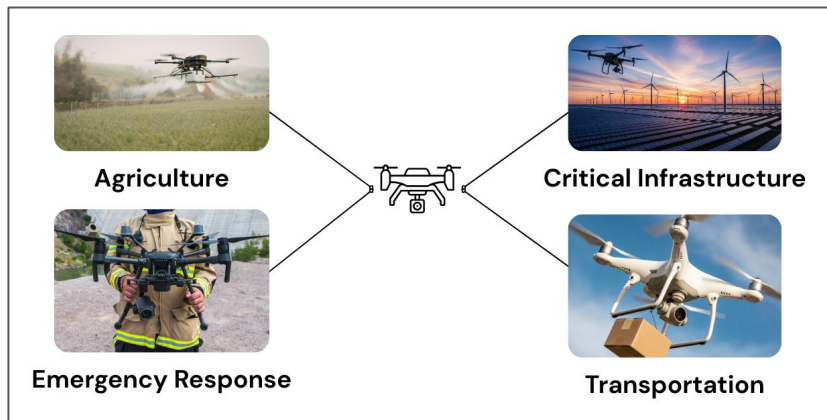


Securing DJI Drones: A Real-Time Wi-Fi Intrusion Detection and Response System

Joseph Park

Ardsley High School, NY

Introduction: Drone Security



Drone usage exploding in recent years

- Versatile, cheap, easy-to-use, powerful

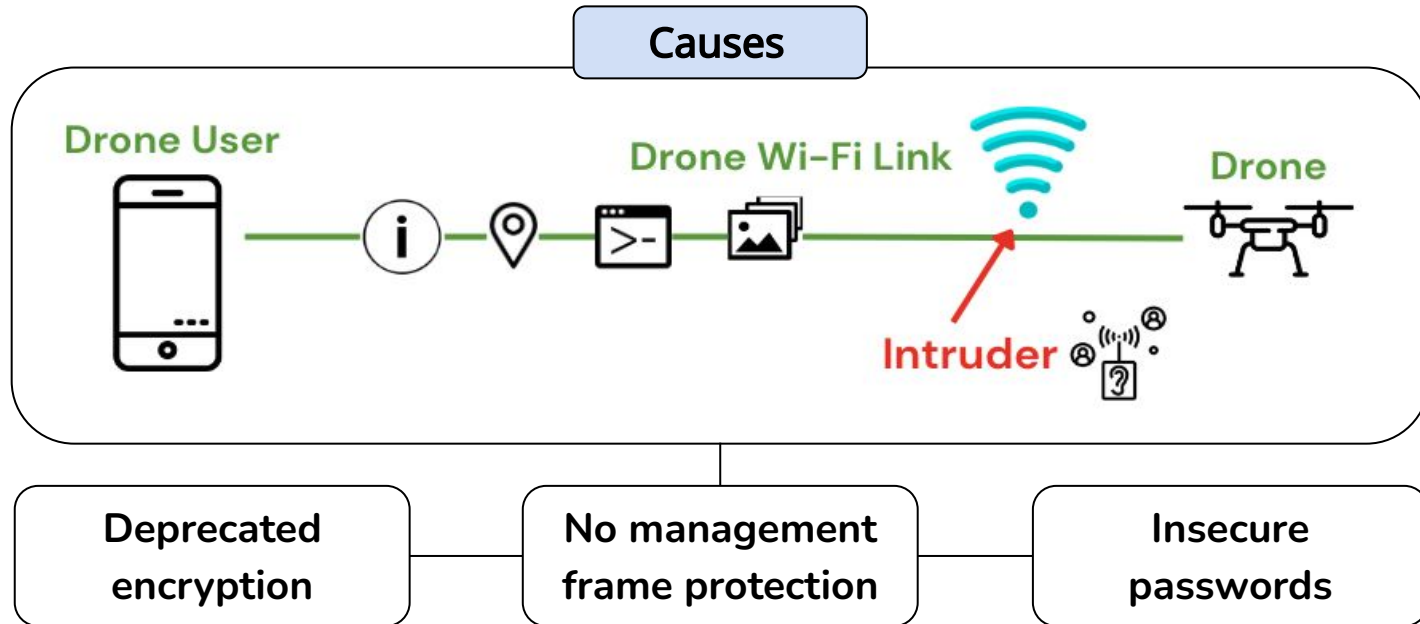
Systematic issues in drone security

- Vendors prioritize low costs
- Simplicity of development/production

Introduction: Wi-Fi Vulnerabilities

Many consumer drones **controlled by a Wi-Fi link**

Wi-Fi security poor; **critical impact**



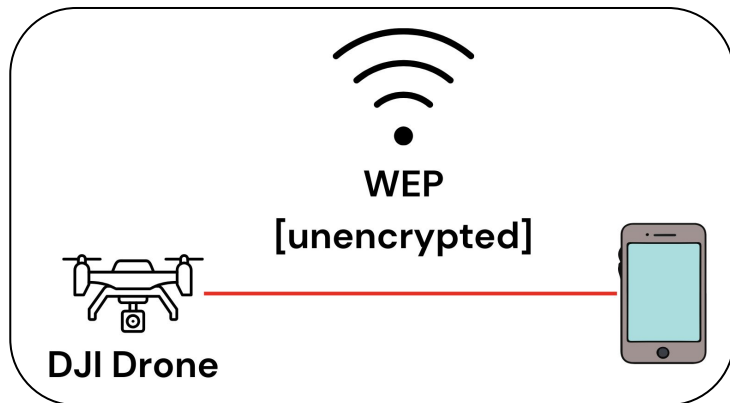
Introduction: DJI Drones

DJI holds ~**70-80%** of the global drone market
(Berg Insight, 2025)

Wi-Fi vulnerabilities common

Deprecated Encryption

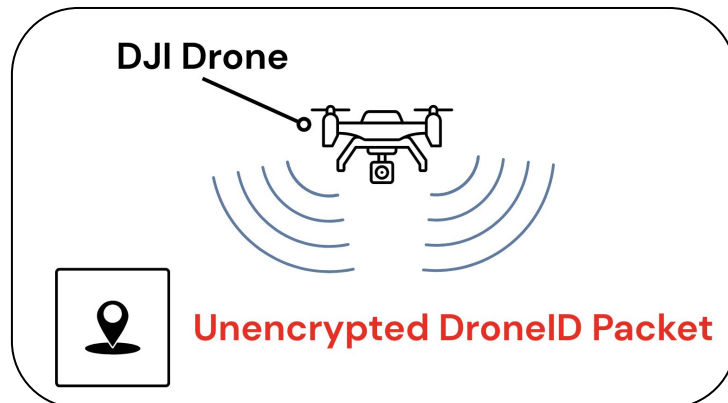
(Pratama et al., 2023)



- DJI Wi-Fi link uses **WEP** protocol
- **Obsolete for decades**

Operator Privacy

(Schiller et al., 2023)



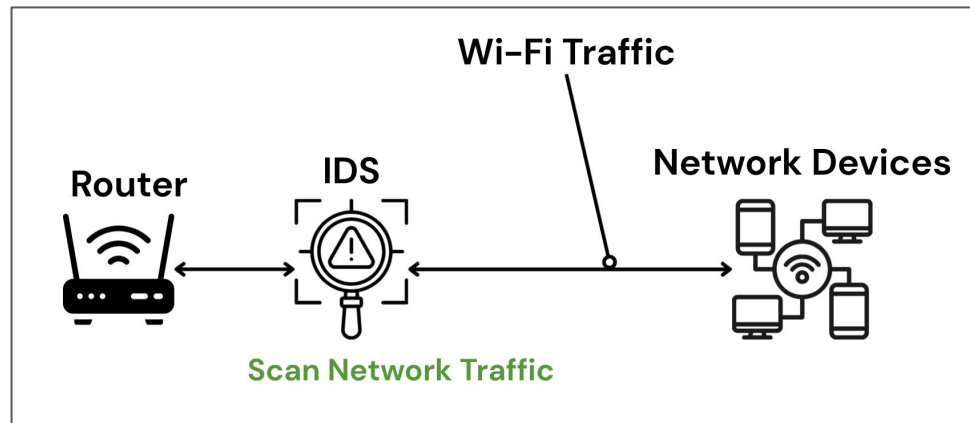
- OcuSync protocol lacks encryption
- Insecure data exposed user location

Introduction: Intrusion Detection Systems

Wi-Fi Network Defense

Intrusion Detection System (IDS/IPS) framework commonly used

Promising use in drone security



Cyber-Physical IDS for UAVs

(Hassler et al., 2023)

- Combines physical and network features
- Fusion of features **found successful** but **no live testing, hardware requirements**

Adversarial, GAN-based GPS IDS

(Asif et al., 2023)

- IDS for **GPS spoofing** & an adversarial attacker
- **93.8% accuracy** but **no live testing**

Gap: Defensive Research in Drones

**Systematic issues &
vulnerabilities in drone Wi-Fi
networks**

**Existing defensive studies
limited in testing methodologies**

Problems // Goals

Problem

Consumer drones (especially DJI, 70% market share), many of which are insecure, lack **real-time, live-tested** defensive systems for Wi-Fi attacks

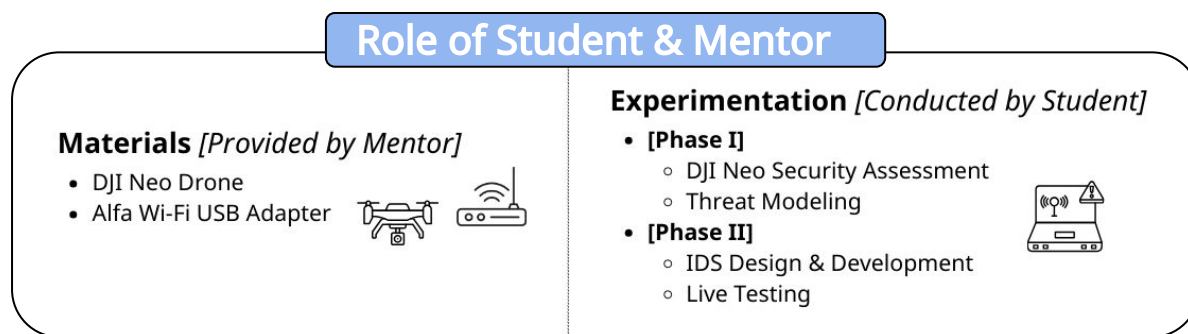
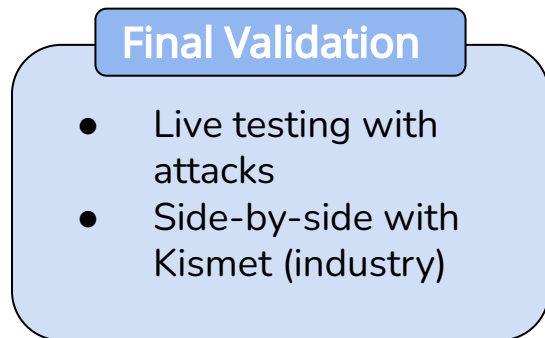
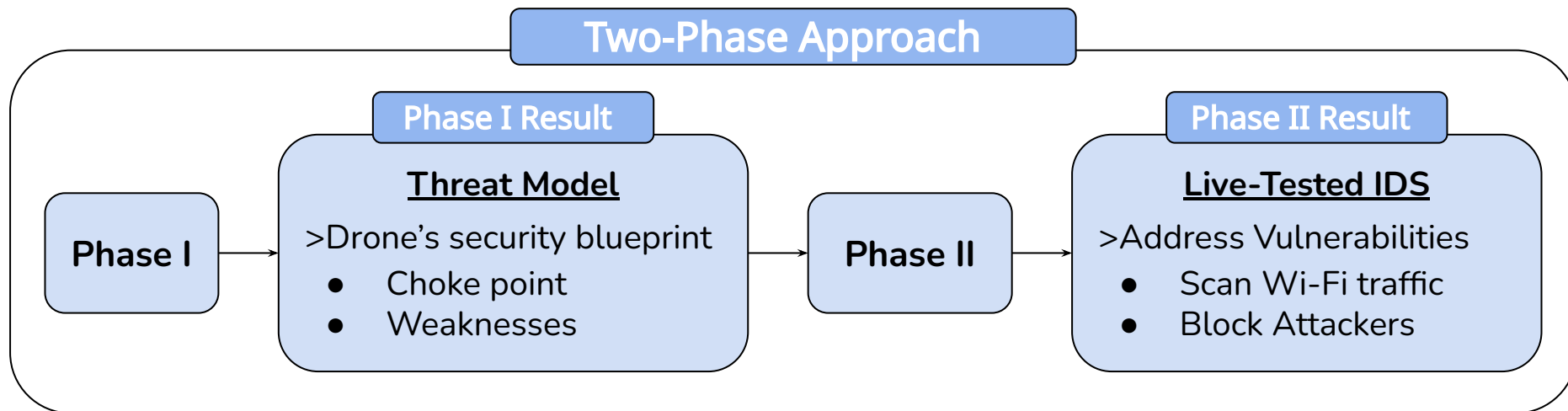
Goal I

Create a threat model of a DJI drone using **penetration testing** and **vulnerability analysis**

Goal II

Develop an **IDS that detects and mitigates Wi-Fi attacks on the DJI drone** using the threat model from Goal I

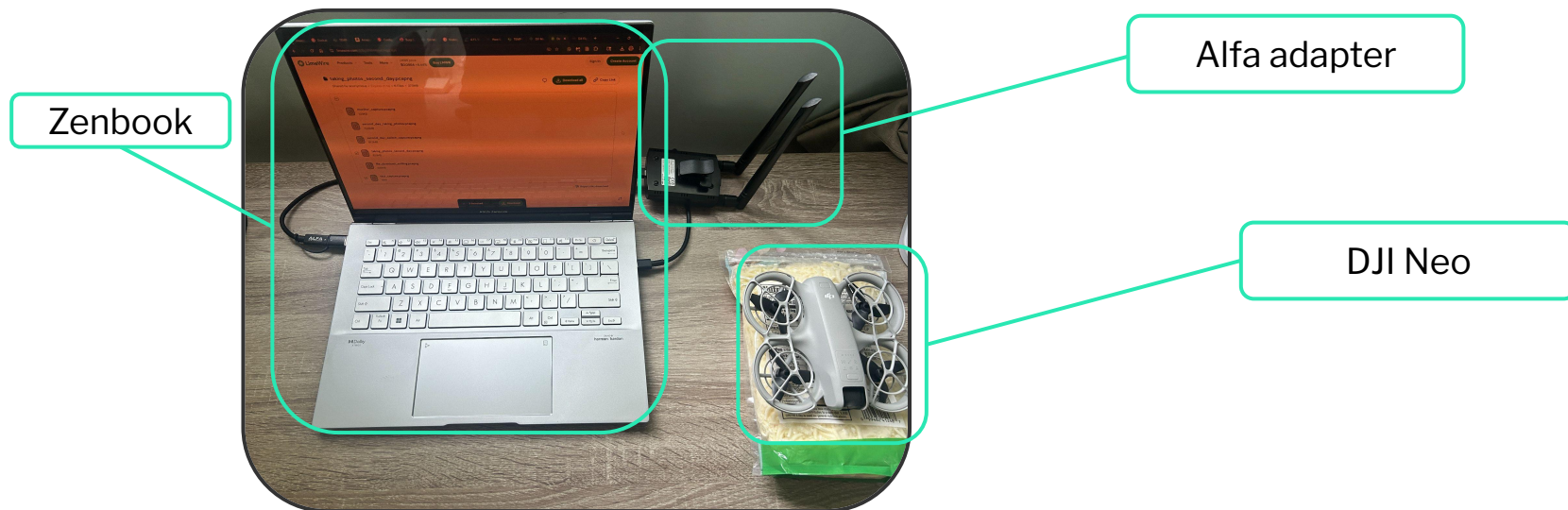
Methods: Two-Phase Approach



Methods: Experimental Testbed

For all wireless testing, an **ASUS Zenbook** equipped via USB with an **Alfa WLAN adapter** was used to join and interact with the Neo's Wi-Fi network.

For penetration testing and live testing, a Kali Linux VM running on VMware Workstation 17 Pro was used to conduct attacks



Methods: Phase I - Vulnerability Assessment



Wi-Fi Pentesting

Common Vulnerability Scoring System v4.0



Vulnerability Analysis

Result: Threat Model

Methods: Phase II - IDS Design & Development

Phase 1, DJI Neo Threat Model



IDS System Architecture

Step 1: Traffic Capture



- Capture Wi-Fi traffic (tshark, python)
- 1-second rolling windows

Step 2: Pattern Detection



- Check for suspicious patterns
- Rules-based detection

Step 3: Alert Operator



- Alert drone user of attack
- Log details for future investigation

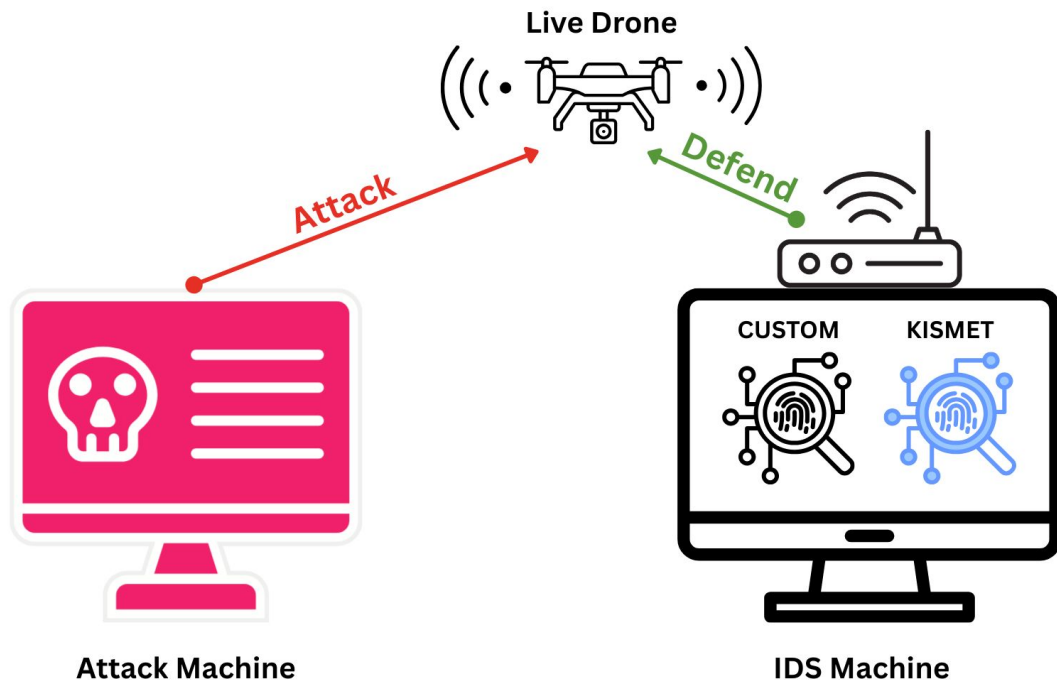
Step 4: Mitigation & Defense



- Block attacking hosts
- Deauthenticate suspicious MACs

Methods: Phase II - Live Evaluation

Live Evaluation Setup



Attack Trials

- Deauthentication attack (5 trials)
- Attacker Connection (5 trials)
- Combined Attack (3 trials)

Evaluation Metrics (vs. Kismet)

- Detection accuracy
- Detection speed
- CPU & RAM consumption

Results: Phase I - Vulnerability Analysis & Threat Model

Table 2 CVSS v4.0 vulnerability assessment of DJI Neo Wi-Fi security findings

Vulnerability	Score	Severity	CVSS v4.0 Vector String
Deauthentication Attack	7.1	HIGH	CVSS:4.0/AV:A/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N
Service Exploitation	5.9	MEDIUM	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:P/VC:H/VI:N/VA:N/SC:N/SI:N/SA:N
ARP Spoofing	4.6	MEDIUM	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:N/VC:L/VI:N/VA:N/SC:N/SI:N/SA:N
Denial-of-Service (TCP)	3.5	LOW	CVSS:4.0/AV:A/AC:L/AT:N/PR:L/UI:P/VC:N/VI:N/VA:L/SC:N/SI:N/SA:N

Note. Two informational findings, Credential Leakage and Open Port, were identified but had insufficient impacts to be classified as exploitable vulnerabilities.

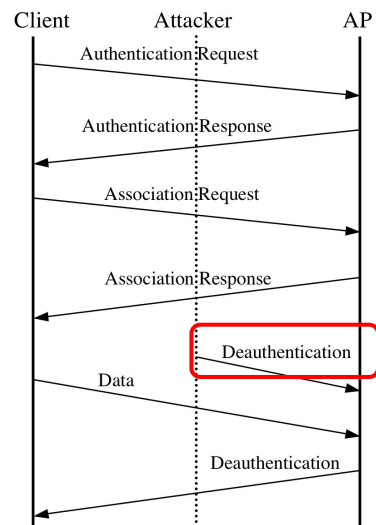
Penetration Testing:
4 vulnerabilities found



Wi-Fi authentication required
for 3 of 4 vulnerabilities



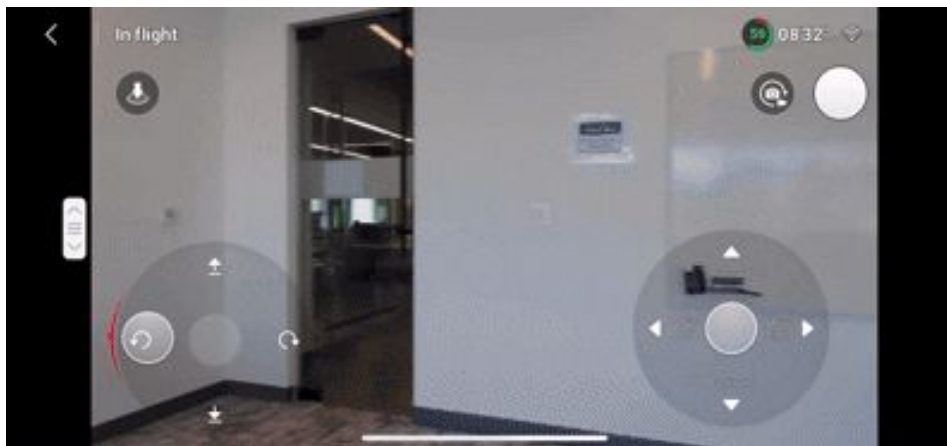
Results: Phase I - Deauth Attack Vulnerability



Attack: Spoof Deauth Request

```
(jospar@kali)-[~]
$ sudo aireplay-ng --deauth 10 -a 58:B8:58:43:0C:89 -c BE:F4:87:DD:A0:C9 wlan0mon
11:13:20 Waiting for beacon frame (BSSID: 58:B8:58:43:0C:89) on channel 153
11:13:20 Sending 64 directed DeAuth (code 7). STMAC: [BE:F4:87:DD:A0:C9] [32|63 ACKs]
11:13:21 Sending 64 directed DeAuth (code 7). STMAC: [BE:F4:87:DD:A0:C9] [43|59 ACKs]
11:13:22 Sending 64 directed DeAuth (code 7). STMAC: [BE:F4:87:DD:A0:C9] [13|57 ACKs]
11:13:22 Sending 64 directed DeAuth (code 7). STMAC: [BE:F4:87:DD:A0:C9] [ 0|55 ACKs]
```

4: Using aireplay-ng, broadcast spoofed deauth packets to drone



DJI FLY app screen capture (controller view after deauth attack)

Results: Phase I - QuickTransfer Vulnerability

QuickTransfer : DJI's file transfer feature to download media from a drone

- Reverse engineer using Wireshark:

1. TCP Handshake

192.168.2.207	192.168.2.1	TCP	78 56694 → 80 [SYN, ECE, CWR]
192.168.2.127	192.168.2.1	TCP	78 56694 → 80 [SYN, ECE, CWR]
192.168.2.1	192.168.2.127	TCP	74 80 → 56694 [SYN, ACK, ECE]
192.168.2.1	192.168.2.207	TCP	74 80 → 56694 [SYN, ACK, ECE]
192.168.2.207	192.168.2.1	TCP	66 56694 → 80 [ACK] Seq=1 Ack=
192.168.2.127	192.168.2.1	TCP	66 56694 → 80 [ACK] Seq=1 Ack=

2. Phone: **GET Request**

Requires no authentication

```
GET /v2?storage=1&path=DCIM/DJI_001/DJI_20250718125917_0010_D.jpg HTTP/1.1
Host: 192.168.2.1
Accept: */*
Content-Type: text/plain
Range: bytes=7720-15526
```

3. Drone: **HTTP Response**

```
HTTP/1.1 206 Partial Content
Content-Type: image/jpeg
Accept-Ranges: bytes
ETag: "1966455292"
Last-Modified: Fri, 18 Jul 2025 16:59:17 GMT
Content-Range: bytes 7720-15526/4898816
```


Results: Phase I - QuickTransfer Vulnerability

No authentication or user verification for file transfer

- Exploit via Wireshark & BurpSuite

▼ Hypertext Transfer Protocol

▼ GET /v2?storage=1&path=DCIM/DJI_001/DJI_20250718125917_0010_D.jpg HTTP/1.1\r\n

Request Method: GET

▼ Request URI: /v2?storage=1&path=DCIM/DJI_001/DJI_20250718125917_0010_D.jpg

Request URI Path: /v2

▼ Request URI Query: storage=1&path=DCIM/DJI_001/DJI_20250718125917_0010_D.jpg

Request URI Query Parameter: storage=1

Request URI Query Parameter: path=DCIM/DJI_001/DJI_20250718125917_0010_D.jpg

Request Version: HTTP/1.1

1: Intercept/bruteforce filepath using Wireshark to sniff

3: Analyze image data - includes GPS metadata

2: Replay GET request via Burp Suite

```
Response
Pretty Raw Hex Render
1 HTTP/1.1 200 OK
2 Content-Type: image/jpeg
3 Accept-Ranges: bytes
4 ETag: "2935358883"
5 Last-Modified: Wed, 16 Jul 2025 17:57:15 GMT
6 Content-Length: 5152768
7 Connection: close
8 Date: Sun, 27 Jul 2025 00:49:06 GMT
9 Server: lighttpd/1.4.55
10
11 y0yā<ExifII*DJI "(1_2Äi60X¶d1(w)t|"020230
```

```
ap/1.0/<?packet begin=..." id="WSM0MpCehiHzreSzNTczkc9d"?>
cx:xmpmeta xmlns:x="adobe:meta/"
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
<rdf:Description rdf:about="DJI Meta Data"
xmlns:tiff="http://ns.adobe.com/tiff/1.0/"
xmlns:exif="http://ns.adobe.com/exif/1.0/"
xmlns:xmp="http://ns.adobe.com/xap/1.0/"
xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/"
dc="http://purl.org/dc/elements/1.1/"
crs="http://ns.adobe.com/camera-raw-settings/1.0/"
drone-dji="http://www.dji.com/drone-dji/1.0/"
xmlns:GPano="http://ns.google.com/photos/1.0/panorama/"
xmlns:Camera="http://pix4d.com/camera/1.0"
xmp:ModifyDate="2025-07-18T12:59:39-04:00"
xmp:CreateDate="2025-07-18T12:59:39-04:00"
tiff:Make="DJI"
tiff:Model="FC8671"
dc:format="image/jpeg"
drone-dji:Version="1.6"
drone-dji:GPSStatus="Normal"
drone-dji:AltitudeType="GeocentricAlt"
drone-dji:GPSLatitude="+41.023829199"
drone-dji:GPSLongitude="-73.836427501"
drone-dji:AbsoluteAltitude="+71.751"
```

GPS coordinates

Wireshark Screen Capture

Results: Phase II: IDS Performance

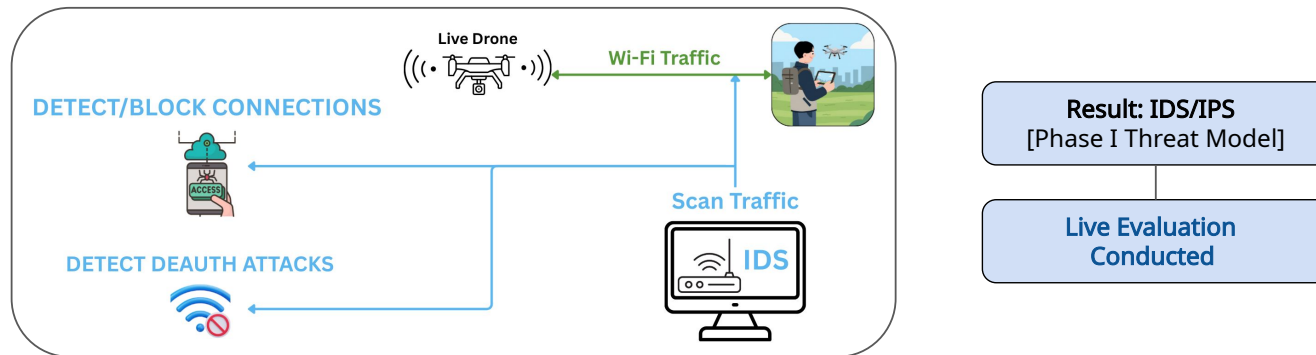


Table 3 Detection accuracy comparison between custom IDRS and Kismet

Attack Type	Trials	IDRS Accuracy (%)	Kismet Accuracy (%)	IDRS Mitigation Rate (%)
Unauth. Connection	5	100.00	100.00	100.00
Deauthentication	5	100.00	100.00	N/A
Combined Attack	3	100.00	100.00	100.00
Overall	13	100.00	100.00	N/A

Note. Both systems successfully detected all 13 attack trials with no false negatives. The IDRS Mitigation rate metric only applied to attempted unauthorized Wi-Fi connections.

Results: Phase II: Supplementary Metrics

Table 4 Detection latency comparison across attack types

Attack Type	Trials	IDRS Mean (s)	IDRS SD (s)	Kismet Mean (s)	Kismet SD (s)
Unauth. Connection	5	2.214	0.78	1.200	0.34
Deauthentication	5	2.370	0.79	1.800	0.51
Combined Attack	3	5.385	2.93	3.000	3.24
Overall Mean	13	3.006	—	1.846	—

Kismet ~1.15 seconds faster on average

Table 5 CPU and Memory utilization statistics

Metric	CPU Usage (%)		Memory Usage (%)	
	IDRS	Kismet	IDRS	Kismet
Mean	21.29	32.45	5.90	1.50
Median	26.00	22.00	6.06	1.75
95th Percentile	34.00	82.34	6.19	2.07
Maximum	46.00	124.00	6.25	2.13

Discussion: Phase II IDRS Comparison

Study IDRS vs. Kismet

Accuracy: Both systems have equal detection accuracy

Latency : Kismet faster by about 1.15~ seconds on average

Overhead:

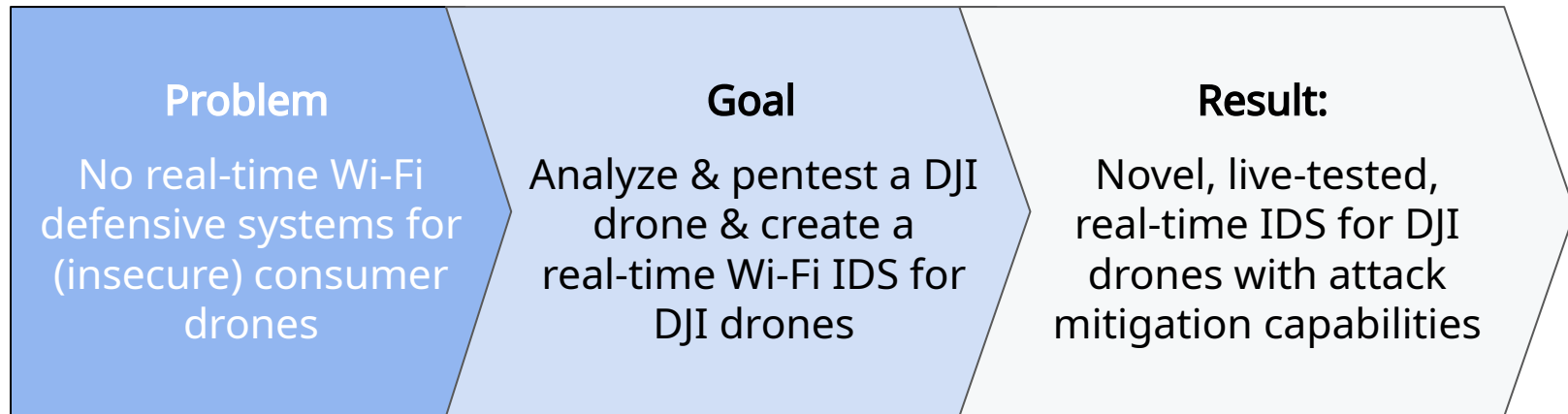
- Custom IDS has steadier, **lower CPU usage**
- Kismet IDS has **lower memory usage**

Custom IDRS - Active Response Capability

Automatically detects unauthorized connections to the drone's Wi-Fi network and blocks them

Conclusion

- Only 1 drone model & environment tested
 - Analyze and test more DJI drone models within the IDS in outdoor environments
- Single-drone system; fleets not evaluated
 - Add IDS support for fleets of drones



Acknowledgements

I would like to thank Dr. Matthew Jablonski and Jad Zarzour for their help - this research would not be possible without their great help!

References

- Abro, G., Zulkifli, S., Masood, R., Asirvadam, V., & Laouiti, A. (2022). Comprehensive review of UAV detection, security, and communication advancements to prevent threats. *Drones*, 6(10), 284. <https://doi.org/10.3390/drones6100284>
- Asif, M., Rahman, M. A., Akkaya, K., Shahriar, H., & Cuzzocrea, A. (2023). Adversarial Data-Augmented Resilient intrusion detection System for unmanned aerial vehicles. *2021 IEEE International Conference on Big Data (Big Data)*, 5428–5437. <https://doi.org/10.1109/bigdata59044.2023.10386140>
- Bayrak, S. (2024). Unveiling intrusions: explainable SVM approaches for addressing encrypted Wi-Fi traffic in UAV networks. *Knowledge and Information Systems*, 66(11), 6675–6695. <https://doi.org/10.1007/s10115-024-02181-9>
- Berg Insight. (2025, April). *Connected Commercial Drones* (1st ed.) Berg Insight. <https://media.berginsight.com/2025/04/02171051/bi-drones1-ps.pdf>
- Grand View Research, Inc. (n.d.). *Drone Market Size, Share & Growth | Industry Report, 2030*. <https://www.grandviewresearch.com/industry-analysis/drone-market-report/toc>
- Hassler, S. C., Mughal, U. A., & Ismail, M. (2023). Cyber-Physical Intrusion Detection System for unmanned aerial vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 25(6), 6106–6117. <https://doi.org/10.1109/tits.2023.3339728>
- Kulp, P., & Mei, N. (2020). A framework for sensing radio frequency spectrum attacks on medical delivery drones. *2022 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 408–413. <https://doi.org/10.1109/smc42975.2020.9283478>
- Mekdad, Y., Aris, A., Babun, L., El Fergougui, A., Conti, M., Lazeretti, R., & Uluagac, A. S. (2023). A survey on security and privacy issues of UAVs. *Computer Networks*, 224, 109626. <https://doi.org/10.1016/j.comnet.2023.109626>
- Mitchell, R., & Chen, I. R. (2014). Adaptive intrusion detection of malicious unmanned air vehicles using behavior rule specifications. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 44(5), 593–604. <https://doi.org/10.1109/TSMC.2013.2265083>
- Pratama, D., Moon, J., Laksmono, A. M. A., Yun, D., Muhammad, I., Jeong, B., Ji, J., & Kim, H. (2023). Behind the wings: the case of reverse engineering and drone hijacking in DJI enhanced Wi-Fi protocol. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2309.05913> m
- Roy, S., Sankaran, S., & Zeng, M. (2024). Green intrusion detection systems: A comprehensive review and directions. *Sensors*, 24(17), 5516. <https://doi.org/10.3390/s24175516>
- Schiller, N., Chlosta, M., Schloegel, M., Bars, N., Eisenhofer, T., Schamowski, T., Domke, F., Schönherr, L., & Holz, T. (2023). Drone Security and the Mysterious Case of DJI's DroneID. *2023 Network and Distributed System Security Symposium*. <https://doi.org/10.14722/ndss.2023.24217>
- Shah, S. F. A., Mazhar, T., Al Shloul, T., Shahzad, T., Hu, Y. C., Mallek, F., & Hamam, H. (2024). Applications, challenges, and solutions of unmanned aerial vehicles in smart city using blockchain. *PeerJ Computer Science*, 10. <https://doi.org/10.7717/peerj-cs.1776>
- Shakhathreh, H., Sawalmeh, A. H., Al-Fuqaha, A., Dou, Z., Almaita, E., Khalil, I., Othman, N. S., Khreishah, A., & Guizani, M. (2019). Unmanned Aerial Vehicles (UAVs): A survey on civil applications and key research challenges. *IEEE Access*, 7, 48572–48634. <https://doi.org/10.1109/access.2019.2909530>
- Tufekci, B., & Tunc, C. (2021). Vulnerability and Threat Analysis of UAVs. *2021 IEEE/ACS 18th International Conference on Computer Systems and Applications (AICCSA)*. <https://doi.org/10.1109/aiccsa53542.2021.9686826>
- Westerlund, O., & Asif, R. (2019). Drone Hacking with Raspberry-Pi 3 and WiFi Pineapple: Security and Privacy Threats for the Internet-of-Things. *2019 1st International Conference on Unmanned Vehicle Systems-Oman (UVS)*, 1. <https://doi.org/10.1109/uvs.2019.8658279>
- Yaacoub, J., Noura, H., Salman, O., & Chehab, A. (2020). Security analysis of drones systems: Attacks, limitations, and recommendations. *Internet of Things*, 11, 100218. <https://doi.org/10.1016/j.iot.2020.100218>

Securing DJI Drones: A Real-Time Wi-Fi Intrusion Detection and Response System

Joseph Park